Question:

An isosceles right triangle with legs of length 8 is partitioned into 16 congruent triangles as shown. The shaded area is

$\text{(A)}\ 10 \qquad \text{(B)}\ 20 \qquad \text{(C)}\ 32 \qquad \text{(D)}\ 40 \qquad \text{(E)}\ 64$

Reasoning + Answer:

<think>

1. First, we need to understand the structure of the partition. The isosceles right triangle is divided into 16 congruent triangles. This means each of these 16 triangles is a small part of the larger triangle.

2. Next, let's determine the area of the large isosceles right triangle. The legs of the isosceles right triangle are given as 8 units each. The area of a right triangle is given by the formula:

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

In this case, the base and height are both 8 units. So, the area of the large triangle is:

$$\text{Area} = \frac{1}{2} \times 8 \times 8 = 32 \text{ square units}$$

3. Since the triangle is partitioned into 16 congruent triangles, we can find the area of each smaller triangle by dividing the area of the large triangle by 16:

$$\text{Area of each smaller triangle} = \frac{32}{16} = 2 \text{ square units}$$

4. Now, we need to count the number of shaded triangles. Observing the diagram, we see that there are 10 shaded triangles.

5. Finally, to find the shaded area, we multiply the area of each smaller triangle by the number of shaded triangles:

$$\text{Shaded area} = 10 \times 2 = 20 \text{ square units}$$

Thus, the shaded area is 20 square units.

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<answer>20</answer>